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Standard Operating Procedure: Building a VMware Cluster with vCenter Server

Title	Build a VMware Cluster with vCenter Server for Electric Petrol
Purpose	To provide a structured step-by-step process in building a VMWare Cluster with three or more servers working with vCenter Server.
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Superseded Documents	N/A
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Revision History

Version	Date	Author	Changes
1.0	29 May 2025	Philip Stewart	Initial Release

Summary of the Policy/Purpose of this SOP

This SOP outlines the process for creating a VMware cluster consisting of three ESXi hosts and a vCenter Server instance using VMware Workstation. The procedure ensures standardized deployment for testing purposes before implementation in production environments.

Scope

This Standard Operating Procedure applies to technicians creating test VMware clusters for electric-petrol's first data centre implementation. The same build instructions apply regardless of the final production cluster size.

Individuals involved

Role	**Responsibility**
Technician	Execute the build process
Supervisor	Approve the final configuration
Network Team	Provide IP addressing and connectivity

Resources and Prerequisites

1. Software:
    - VMware Workstation 16/17 Pro
    - ESXi 6.7/7.0 installation ISO
    - vCenter Server Appliance (VCSA) ISO
2. Recommended Hardware Requirements:
    - Host machine with a minimum of 32GB RAM
    - 200GB of free disk space
    - Intel VT-x/AMD-V enabled in the BIOS
3. Network Requirements:

Location	Subnet	Purpose	Default Gateway	DNS/NTP Servers	Domain
Dublin Head Office	10.1.0.0/16	Core Production	10.1.1.1	10.1.6.10 (DNS), 10.1.6.20 (NTP)	HeadOffice.Electric-petrol.ie
Balbriggan Office	10.32.0.0/16	DR/Secondary Prod	10.32.1.1	10.32.6.10 (DNS), 10.32.6.20 (NTP)	Balbriggan.Electric-petrol.ie
Cork Data Centre	10.2.0.0/16	Test/Dev Cluster	10.2.1.1	10.2.6.10 (DNS), 10.2.6.20 (NTP)	Datacentre1.electric-petrol.ie

VMWare Cluster IP Allocations

Dublin Head Office (10.1.0.0/16)

- **ESXi Management**: 10.1.10.10 - 10.1.10.20
- **vCenter Server**: 10.1.10.30
- **vMotion**: 10.1.20.0/24
- **Production Virtual Machine Network**: 10.1.30.0/24

Balbriggan Office (10.32.0.0/16)

- **ESXi Management**: 10.32.10.10 - 10.32.10.20
- **vCenter Server**: 10.32.10.30
- **Disaster Recovery vMotion**: 10.32.20.0/24

- **Disaster Recovery:** 10.32.30.0/24

Cork Data Centre (10.2.0.0/16)

- **ESXi Management**: 10.2.10.10 - 10.2.10.20
- **vCenter Server**: 10.2.10.30
- **Test vMotion**: 10.2.20.0/24
- **Test Virtual Machine Network**: 10.2.30.0/24
- Private network segment for management
- Static IP addresses are allocated for:
  - 3 ESXi hosts
  - vCenter Server
  - DNS server (if applicable)

4. Naming Convention To Be Used:
   - ESXi hosts: ESXi-*Country*-*Site*-*number*
   - vCenter: VC-*Country**Dev, Test, or DR*

Procedure

1. Prepare the Environment

1.1 Create a new custom virtual network in VMware Workstation:

- Edit > Virtual Network Editor
- Add network (VMnet2 recommended)
- Configure as Host-only with DHCP disabled

1.2 Verify all required ISO files are downloaded and checksums validated

2. Deploy ESXi Hosts

2.1 Create first ESXi VM:

- New Virtual Machine > Custom
- Hardware compatibility: ESXi 6.7/7.0
- 4 vCPUs, 8GB RAM, 40GB disk
- Network adapter: VMnet2
- Install ESXi using the ISO

- Set root password and configure management network

2.2 Repeat for two additional ESXi hosts with unique hostnames/IPs

2.3 Verify all hosts are accessible via SSH and web interface

3. Deploy vCenter Server

3.1 Mount VCSA ISO and run the installer

3.2 Select “Install < vCenter Server’ (embedded deployment).

3.3 Configure

- System name:VC-IRE-*Dev, Test, or DR*
- Root password
- Deployment size: "Tiny" (for testing)
- Datastore: Auto-select
- Network settings: Static IP on VMnet2
  3.4 Complete first stage deployment
  3.5 Access https://[vCenterIP]:5480 to complete setup
  3.6 Configure SSO domain (vsphere.local recommended)

4. Create VMware Cluster

4.1 Log in to vCenter Web Client

4.2 Create new Datacenter

4.3 Create new cluster

- Enable DRS (set this to manual mode)
- Enable High Availability (HA) (optional for testing)
  4.4 Add all three ESXi hosts to the cluster
  4.5 Verify all hosts show as connected

5. Configure Storage

5.1 For each host

- Add a shared datastore (NFS/iSCSI if testing shared storage)
- You can also configure local storage for testing if you wish

6. Network Configuration

6.1 Create standard switch or distributed switch

6.2 Configure Port Groups:

- Management

- VM Network
- vMotion (if applicable)

7. Final Configuration

- Set up NTP on all hosts pointing to same time source
- Configure syslog if required
- Apply any organisational security policies

Acceptance Tests

1. Verify that all ESXi hosts are visible in vCenter
2. Create a test VM on each host
3. Verify that vMotion works between hosts (if configured)
4. Test the High Availability by forcibly powering off one host
5. Verify backup system can access all components

Output

1. Document all IP addresses/hostnames in CMDB
2. Create build book entry with:
    - Cluster specifications
    - Network diagram
    - Any deviations from standard
3. Handover documentation to operations team

Appendices

Appendix A: Troubleshooting

- If hosts fail to connect to vCenter:
    1. Verify network connectivity
    2. Check DNS resolution
    3. Verify NTP synchronization
    4. Restart management agents on ESXi
- Common installation errors:

- Ensure hardware virtualization is enabled in BIOS
- Verify sufficient RAM on host machine
- Check for conflicting network configurations

Appendix B: Common Issues and Fix scripts

This appendix provides some common failure scenarios and how best to resolve the issues using PowerCLI or ESXi shell commands.

Issue	Symptom	Resolution Command
Host Not Visible in vCenter	Host Shows as disconnected	Services.sh restart on the ESXi host
DNS issues	Hostnames not resolving correctly	Add entries to /etc/hosts or verify the DNS configuration
NTP is not Syncing	Clock drifting between hosts	Esxcli hardware clock get/set

Appendix C: Cross-Site Connectivity & Failover Procedures

1. Site-to-Site Communication Requirements

Type of Traffic	Source Subnet	Destination Subnet	Purpose
vCenter Replication	10.32.10.0/24 (DR)	10.1.10.0/24 (Prod)	Balbriggan DR – Dublin Head Office
Backup Traffic	10.2.19.0/24 (Dev/test)	10.1.40.0/24	Cork Dev/test – Dublin backup NAS
NTP sync	All sites	10.1.6.20 (Dublin)	Time Synchronisation

2. Failover Scenarios

- Scenario 1: Outage in Dublin.
  A) Activate DR cluster in Balbriggan:
    - Power on critical VMs from backups in the Balbriggan NAS
    - Update DNS to point to Balbriggan IP addresses in the 10.32.0.0/24 range

  B) Network Redirect:
    - Re-routing all traffic from Dublins subnet (10.1.0.0/16) to Balbriggan (10.32.0.0/16) using firewall rules
- Scenario 2: Test/Dev Isolation in Cork

A) Block accidental production writes by having cork block ip addresses from the production environment and whitelist desired Ips